

Amyotrophic Lateral Sclerosis (ALS): Pathophysiology and Revised Diagnostic Criteria

International Federation of Neurology

BACKGROUND: Amyotrophic lateral sclerosis (ALS) is a fatal neurodegenerative disorder characterized by the progressive loss of upper and lower motor neurons in the brain and spinal cord.

PATHOPHYSIOLOGY: Approximately 10% of cases are familial, linked to mutations in SOD1, C9orf72, TARDBP, and FUS genes. The exact mechanisms involve glutamate excitotoxicity, oxidative stress, and mitochondrial dysfunction.

CLINICAL FEATURES: Patients typically present with asymmetric limb weakness, muscle atrophy, fasciculations, and eventually bulbar symptoms such as dysphagia and respiratory failure. Mean survival is 3 to 5 years from symptom onset.